

高等微积分 (2)

第十四章：多变量函数的微分学

习题课

提要： 典型例题等

综合练习

例 19. 设曲面 S 由 $ax + by + cz = G(x^2 + y^2 + z^2)$ 确定, 其中 G 可导并且 a, b, c 不全为零. 求证: S 上任意点处的法线与某定直线相交或平行.

证明: 令 $F(x, y, z) = G(x^2 + y^2 + z^2) - (ax + by + cz)$, 则

$$\text{grad} F = \begin{pmatrix} 2xG'(x^2 + y^2 + z^2) - a \\ 2yG'(x^2 + y^2 + z^2) - b \\ 2zG'(x^2 + y^2 + z^2) - c \end{pmatrix}.$$

设所求定直线的方程为 $\frac{x-x_1}{\alpha} = \frac{y-y_1}{\beta} = \frac{z-z_1}{\gamma}$.

上述直线与曲面 S 上任意点 $P_0(x_0, y_0, z_0)$ 处的法线相交或平行当且仅当下述三个向量

$$\operatorname{grad} F(P_0), \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}, \begin{pmatrix} x_1 - x_0 \\ y_1 - y_0 \\ z_1 - z_0 \end{pmatrix}$$

线性相关, 也即我们有

$$\begin{vmatrix} 2x_0 G'(x_0^2 + y_0^2 + z_0^2) - a & \alpha & x_1 - x_0 \\ 2y_0 G'(x_0^2 + y_0^2 + z_0^2) - b & \beta & y_1 - y_0 \\ 2z_0 G'(x_0^2 + y_0^2 + z_0^2) - c & \gamma & z_1 - z_0 \end{vmatrix} = 0.$$

根据行列式的性质, 该式等价于

$$2G'(x_0^2 + y_0^2 + z_0^2) \begin{vmatrix} x_0 & \alpha & x_1 - x_0 \\ y_0 & \beta & y_1 - y_0 \\ z_0 & \gamma & z_1 - z_0 \end{vmatrix} = \begin{vmatrix} a & \alpha & x_1 - x_0 \\ b & \beta & y_1 - y_0 \\ c & \gamma & z_1 - z_0 \end{vmatrix}.$$

而当 $\alpha = a, \beta = b, \gamma = c, x_1 = y_1 = z_1 = 0$ 时,
上式成立, 也即曲面任意点处的法线与定直线

$$\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$$

平行或相交.

例 20. 设 $f: (0, +\infty) \rightarrow \mathbb{R}$ 为二阶连续可导函数
且 $u(x, y, z) = f(\sqrt{x^2 + y^2 + z^2})$ 满足

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0,$$

求函数 f 的表达式.

解: 由题设可知 $\frac{\partial u}{\partial x} = \frac{xf'(\sqrt{x^2+y^2+z^2})}{\sqrt{x^2+y^2+z^2}}$, 则

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{f'(\sqrt{x^2+y^2+z^2})}{\sqrt{x^2+y^2+z^2}} + \frac{x^2 f''(\sqrt{x^2+y^2+z^2})}{x^2+y^2+z^2} \\ &\quad - \frac{x^2 f'(\sqrt{x^2+y^2+z^2})}{(x^2+y^2+z^2)^{\frac{3}{2}}}. \end{aligned}$$

由对称性可得

$$\frac{\partial^2 u}{\partial y^2} = \frac{f'(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}} + \frac{y^2 f''(\sqrt{x^2 + y^2 + z^2})}{x^2 + y^2 + z^2} - \frac{y^2 f'(\sqrt{x^2 + y^2 + z^2})}{(x^2 + y^2 + z^2)^{\frac{3}{2}}},$$

$$\frac{\partial^2 u}{\partial z^2} = \frac{f'(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}} + \frac{z^2 f''(\sqrt{x^2 + y^2 + z^2})}{x^2 + y^2 + z^2} - \frac{z^2 f'(\sqrt{x^2 + y^2 + z^2})}{(x^2 + y^2 + z^2)^{\frac{3}{2}}},$$

于是我们有

$$\begin{aligned} 0 &= \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{f'(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}} + \frac{x^2 f''(\sqrt{x^2 + y^2 + z^2})}{x^2 + y^2 + z^2} \\ &\quad - \frac{x^2 f'(\sqrt{x^2 + y^2 + z^2})}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} + \frac{f'(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}} + \frac{y^2 f''(\sqrt{x^2 + y^2 + z^2})}{x^2 + y^2 + z^2} \\ &\quad - \frac{y^2 f'(\sqrt{x^2 + y^2 + z^2})}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} + \frac{f'(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}} + \frac{z^2 f''(\sqrt{x^2 + y^2 + z^2})}{x^2 + y^2 + z^2} \\ &\quad - \frac{z^2 f'(\sqrt{x^2 + y^2 + z^2})}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} = \frac{2f'(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}} + f''(\sqrt{x^2 + y^2 + z^2}). \end{aligned}$$

从而 $\forall r > 0$, $f''(r) + \frac{2f'(r)}{r} = 0$, 故 $f'(r) = -\frac{C_1}{r^2}$,
进而 $f(r) = \frac{C_1}{r} + C_2$, 其中 C_1, C_2 为常数.

例 21. 设 $u = x^2 + y^2 + z^2$, 其中 $z = z(x, y)$ 是由方程 $z = x - ye^z$ 确定的隐函数, 求 du 和 $\frac{\partial^2 u}{\partial x \partial y}$.

解: 由题设知 $\frac{\partial u}{\partial x} = 2x + 2z \frac{\partial z}{\partial x}$, $\frac{\partial u}{\partial y} = 2y + 2z \frac{\partial z}{\partial y}$.

注意到 $z = x - ye^z$, 两边分别对 x, y 求偏导数可得 $\frac{\partial z}{\partial x} = \frac{1}{1+ye^z}$, $\frac{\partial z}{\partial y} = -\frac{e^z}{1+ye^z}$. 于是

$$du = \left(2x + \frac{2z}{1+ye^z}\right)dx + \left(2y - \frac{2ze^z}{1+ye^z}\right)dy.$$

与此同时, 我们也有

$$\begin{aligned}\frac{\partial^2 u}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(2y + 2z \frac{\partial z}{\partial y} \right) = 2 \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} + 2z \frac{\partial^2 z}{\partial x \partial y} \\&= -\frac{2e^z}{(1 + ye^z)^2} + 2z \frac{\partial}{\partial x} \left(-\frac{e^z}{1 + ye^z} \right) \\&= -\frac{2e^z}{(1 + ye^z)^2} - 2z \cdot \frac{e^z(1 + ye^z) \frac{\partial z}{\partial x} - e^z \cdot ye^z \frac{\partial z}{\partial x}}{(1 + ye^z)^2} \\&= -\frac{2e^z}{(1 + ye^z)^2} - \frac{2ze^z}{(1 + ye^z)^3} \\&= -\frac{2e^z(1 + ye^z + z)}{(1 + ye^z)^3} = -\frac{2(1 + x)e^z}{(1 + ye^z)^3}.\end{aligned}$$

例 22. 假设 $z = z(x, y)$ 为二阶连续可导且满足

$$A \frac{\partial^2 z}{\partial x^2} + 2B \frac{\partial^2 z}{\partial x \partial y} + C \frac{\partial^2 z}{\partial y^2} = 0,$$

其中 $B^2 - AC > 0$ 且 $C \neq 0$. 若令

$$\begin{cases} u = x + \alpha y, \\ v = x + \beta y, \end{cases}$$

试确定 α, β 为何值时能使原方程变为

$$\frac{\partial^2 z}{\partial u \partial v} = 0.$$

解: 由题设可知

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} = \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) z,$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right)^2 z,$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \alpha + \frac{\partial z}{\partial v} \cdot \beta = \left(\alpha \frac{\partial}{\partial u} + \beta \frac{\partial}{\partial v} \right) z,$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = \left(\alpha \frac{\partial}{\partial u} + \beta \frac{\partial}{\partial v} \right)^2 z,$$

$$\begin{aligned} \frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial x} \left(\alpha \frac{\partial}{\partial u} + \beta \frac{\partial}{\partial v} \right) z \\ &= \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left(\alpha \frac{\partial}{\partial u} + \beta \frac{\partial}{\partial v} \right) z. \end{aligned}$$

帶入題設方程可得

$$\begin{aligned} 0 &= A \frac{\partial^2 z}{\partial x^2} + 2B \frac{\partial^2 z}{\partial x \partial y} + C \frac{\partial^2 z}{\partial y^2} \\ &= A \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right)^2 z + 2B \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left(\alpha \frac{\partial}{\partial u} + \beta \frac{\partial}{\partial v} \right) z \\ &\quad + C \left(\alpha \frac{\partial}{\partial u} + \beta \frac{\partial}{\partial v} \right)^2 z \\ &= (A + 2B\alpha + C\alpha^2) \frac{\partial^2 z}{\partial u^2} \\ &\quad + 2(A + B(\alpha + \beta) + C\alpha\beta) \frac{\partial^2 z}{\partial u \partial v} \\ &\quad + (A + 2B\beta + C\beta^2) \frac{\partial^2 z}{\partial v^2}. \end{aligned}$$

于是题设方程能变成 $\frac{\partial^2 z}{\partial u \partial v} = 0$ 当且仅当

$$A + 2B\alpha + C\alpha^2 = 0, \quad A + 2B\beta + C\beta^2 = 0,$$

$$A + B(\alpha + \beta) + C\alpha\beta \neq 0.$$

由于 $B^2 - AC > 0$, 因此我们有

$$\alpha = \frac{-B \mp \sqrt{B^2 - AC}}{C},$$

$$\beta = \frac{-B \pm \sqrt{B^2 - AC}}{C}.$$

此时我们还有

$$\begin{aligned} & A + B(\alpha + \beta) + C\alpha\beta \\ &= A - \frac{2B^2}{C} + A = \frac{2}{C}(AC - B^2) \neq 0. \end{aligned}$$

于是题设方程能变成 $\frac{\partial^2 z}{\partial u \partial v} = 0$ 当且仅当

$$\alpha = \frac{-B \mp \sqrt{B^2 - AC}}{C}, \quad \beta = \frac{-B \pm \sqrt{B^2 - AC}}{C}.$$

此时存在两个连续可导函数 f, g 使得

$$z(x, y) = f(u) + g(v) = f(x + \alpha y) + g(x + \beta y).$$

例 23. 设 $x = f(y, z)$, $y = g(x, z)$, $z = h(x, y)$, 其中 f, g, h 为可微函数. 求证:

$$\frac{\partial f}{\partial y} \cdot \frac{\partial g}{\partial z} \cdot \frac{\partial h}{\partial x} = -1.$$

证明: 由题设可知

$$x = f(g(x, z), z), \quad z = h(x, g(x, z)).$$

由此立刻可得

$$1 = \frac{\partial f}{\partial y} \cdot \frac{\partial g}{\partial x}, \quad 0 = \frac{\partial h}{\partial x} + \frac{\partial h}{\partial y} \cdot \frac{\partial g}{\partial x}, \quad 1 = \frac{\partial h}{\partial y} \cdot \frac{\partial g}{\partial z}.$$

于是我们就有

$$\frac{\partial f}{\partial y} = \frac{1}{\frac{\partial g}{\partial x}}, \quad \frac{\partial h}{\partial x} = -\frac{\partial h}{\partial y} \cdot \frac{\partial g}{\partial x}, \quad \frac{\partial g}{\partial z} \cdot \frac{\partial h}{\partial y} = 1.$$

进而立刻可得知

$$\begin{aligned} \frac{\partial f}{\partial y} \cdot \frac{\partial g}{\partial z} \cdot \frac{\partial h}{\partial x} &= \frac{1}{\frac{\partial g}{\partial x}} \cdot \frac{\partial g}{\partial z} \cdot \left(-\frac{\partial h}{\partial y} \frac{\partial g}{\partial x} \right) \\ &= -\frac{\partial g}{\partial z} \cdot \frac{\partial h}{\partial y} = -1. \end{aligned}$$

例 24. $\forall (x, y, z) \in \mathbb{R}^3$, 定义

$$f(x, y, z) = x^2 + 2y - xyz.$$

设 f 在点 $(1, 1, 0)$ 处的梯度方向为 $\vec{\ell}$, 求 $\frac{\partial f}{\partial \ell}(1, 1, 0)$.

解: 由题设可知

$$\text{grad} f(1, 1, 0) = \begin{pmatrix} 2x - yz \\ 2 - xz \\ -xy \end{pmatrix} \Big|_{(1,1,0)} = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix},$$

于是我们有 $\frac{\partial f}{\partial \ell}(1, 1, 0) = \|\text{grad} f(1, 1, 0)\| = 3$.

例 25. 设隐函数 $z = z(x, y)$ 由方程

$$z = f(x + y + z)$$

确定, 其中 f 为 $\mathcal{C}^{(2)}$ 类函数且 $f' \neq 1$, 求 $\frac{\partial^2 z}{\partial x^2}$.

解: 将方程 $z = f(x + y + z)$ 对 x 求偏导数, 则

$$\begin{aligned}\frac{\partial z}{\partial x} &= f'(x + y + z) \frac{\partial(x + y + z)}{\partial x} \\ &= f'(x + y + z) \left(1 + \frac{\partial z}{\partial x}\right),\end{aligned}$$

由此立刻可得 $\frac{\partial z}{\partial x} = \frac{f'(x+y+z)}{1-f'(x+y+z)}.$

进而我们就有

$$\begin{aligned}\frac{\partial^2 z}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{f'(x+y+z)}{1-f'(x+y+z)} \right) \\&= \frac{\partial}{\partial x} \left(-1 + \frac{1}{1-f'(x+y+z)} \right) \\&= -\frac{1}{(1-f'(x+y+z))^2} \cdot \frac{\partial(1-f'(x+y+z))}{\partial x} \\&= \frac{f''(x+y+z)}{(1-f'(x+y+z))^2} \cdot \frac{\partial(x+y+z)}{\partial x} \\&= \frac{f''(x+y+z)}{(1-f'(x+y+z))^2} \left(1 + \frac{\partial z}{\partial x} \right) \\&= \frac{f''(x+y+z)}{(1-f'(x+y+z))^3}.\end{aligned}$$

例 26. 设 p 为实数. $\forall (x, y) \in \mathbb{R}^2$, 令

$$f(x, y) = \begin{cases} (x^2 + y^2)^p \sin \frac{1}{\sqrt{x^2 + y^2}}, & \text{若 } (x, y) \neq (0, 0), \\ 0, & \text{若 } (x, y) = (0, 0). \end{cases}$$

请分析 p 取何值时, 函数 f 在原点处:

(1) 连续; (2) 可导; (3) 可微.

解: (1) 若 $p > 0$, 则 $\forall (x, y) \in \mathbb{R}^2$, 我们有

$$|f(x, y)| \leq (x^2 + y^2)^p.$$

由夹逼原理可得 $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0 = f(0, 0)$.

因此函数 f 在原点处连续.

现假设 $p \leq 0$. $\forall n \geq 1$, 令 $x_n = \frac{1}{2n\pi}$, $y_n = \frac{1}{2n\pi + \frac{\pi}{2}}$,

则 $f(x_n, 0) = 0$, $f(y_n, 0) = y_n^{2p} \geq 1$. 故 $\{x_n\}$, $\{y_n\}$

收敛到 0, 但 $\{f(x_n, 0)\}$, $\{f(y_n, 0)\}$ 却不收敛到

同一个极限. 这表明 f 在原点间断.

综上所述可知 f 在原点连续当且仅当 $p > 0$.

(2) 若 $p > \frac{1}{2}$, 由夹逼原理可知

$$\lim_{x \rightarrow 0} \frac{f(x,0)-f(0,0)}{x} = \lim_{x \rightarrow 0} |x|^{2p-1}(\operatorname{sgn} x) \sin \frac{1}{|x|} = 0,$$

$$\lim_{y \rightarrow 0} \frac{f(0,y)-f(0,0)}{y} = \lim_{y \rightarrow 0} |y|^{2p-1}(\operatorname{sgn} y) \sin \frac{1}{|y|} = 0,$$

故 f 在原点可导且 $f'_x(0,0) = f'_y(0,0) = 0$.

若 $p \leq \frac{1}{2}$, 由 $\lim_{x \rightarrow 0} (\operatorname{sgn} x) \sin \frac{1}{|x|}$ 不存在可知 f 在原点不可导. 故 f 在原点可导当且仅当 $p > \frac{1}{2}$.

(3) 若 f 在原点可微, 则它在该点可导, 故 $p > \frac{1}{2}$.

现假设 $p > \frac{1}{2}$, 此时我们有

$$\begin{aligned} & \lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0) - f'_x(0,0)x - f'_y(0,0)y}{\sqrt{x^2 + y^2}} \\ &= \lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2)^{p-\frac{1}{2}} \sin \frac{1}{\sqrt{x^2 + y^2}} = 0, \end{aligned}$$

于是由可微的定义可知 f 在原点可微.

综上所述可知 f 在原点可微当且仅当 $p > \frac{1}{2}$.

例 27. 固定 $k > 0$. $\forall (x, y) \in \mathbb{R}^2$, 定义

$$f(x, y) = \begin{cases} \frac{|xy|^k}{x^2+y^2}, & \text{若 } (x, y) \neq (0, 0), \\ 0, & \text{若 } (x, y) = (0, 0). \end{cases}$$

问 k 为何值时函数 f 在原点连续, 可导, 可微或连续可导?

解: 连续性. 当 $k > 1$ 时, $\forall (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$,

$$\frac{|xy|^k}{x^2+y^2} \leq \frac{(\frac{1}{2}(x^2+y^2))^k}{x^2+y^2} = \frac{1}{2^k}(x^2+y^2)^{k-1}.$$

于是由夹逼原理可知此时函数 f 在原点连续.

当 $k = 1$ 时, 由于 $\lim_{x \rightarrow 0} f(x, x) = \frac{1}{2}$, 由复合极限法则可知此时函数 f 在原点不连续.

当 $k < 1$ 时, 因 $\lim_{x \rightarrow 0} f(x, x) = \lim_{x \rightarrow 0} \frac{1}{2}|x|^{2k-2} = +\infty$, 则由复合极限法则可知 f 在原点不连续.

综上所述可知 f 在原点连续当且仅当 $k > 1$.

可导性. $\forall x, y \in \mathbb{R} \setminus \{0\}$, 因 $f(x, 0) = f(0, y) = 0$, 于是由偏导数的定义可知 $\frac{\partial f}{\partial x}(0, 0) = \frac{\partial f}{\partial y}(0, 0) = 0$, 故函数 f 在原点处可导.

可微性. 因 $\frac{\partial f}{\partial x}(0,0) = \frac{\partial f}{\partial y}(0,0) = 0$, 则由微分的定义立刻可知 f 在原点可微当且仅当

$$\lim_{(x,y) \rightarrow (0,0)} \frac{|xy|^k}{(x^2 + y^2)^{\frac{3}{2}}} = 0,$$

而这又等价于说

$$\lim_{(x,y) \rightarrow (0,0)} \frac{|xy|^{\frac{2k}{3}}}{x^2 + y^2} = 0,$$

由此可知 f 在原点处可微当且仅当 $k > \frac{3}{2}$.

连续可导性. 若 f 在原点连续可导, 则由前面讨论可知 $k > \frac{3}{2}$. 此时 $\forall (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$,

$$\frac{\partial f}{\partial x}(x, y) = \frac{k|x|^{k-1}|y|^k \operatorname{sgn} x}{x^2 + y^2} - \frac{2x|xy|^k}{(x^2 + y^2)^2},$$

$$\frac{\partial f}{\partial y}(x, y) = \frac{k|x|^k|y|^{k-1} \operatorname{sgn} y}{x^2 + y^2} - \frac{2y|xy|^k}{(x^2 + y^2)^2}.$$

由此立刻可得

$$\begin{aligned} \left| \frac{\partial f}{\partial x}(x, y) \right| &\leq \frac{k|x|^{k-1}|y|^k}{x^2 + y^2} + \frac{2|x|^{k+1}|y|^k}{(x^2 + y^2)^2} \\ &\leq \frac{k}{2^{k-1}}(x^2 + y^2)^{k-2}|y| + \frac{1}{2^{k-1}}(x^2 + y^2)^{k-2}|x| \\ &\leq \frac{k}{2^{k-1}}(x^2 + y^2)^{k-\frac{3}{2}} + \frac{1}{2^{k-1}}(x^2 + y^2)^{k-\frac{3}{2}} \\ &= \frac{k+1}{2^{k-1}}(x^2 + y^2)^{k-\frac{3}{2}}, \\ \left| \frac{\partial f}{\partial y}(x, y) \right| &\leq \frac{k|x|^k|y|^{k-1}}{x^2 + y^2} + \frac{2|x|^k|y|^{k+1}}{(x^2 + y^2)^2} \leq \frac{k+1}{2^{k-1}}(x^2 + y^2)^{k-\frac{3}{2}}, \end{aligned}$$

由夹逼原理可知 $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ 在原点连续.

例 28. 假设由方程

$$f(u^2 - x^2, u^2 - y^2, u^2 - z^2) = 0$$

可确定 $u = \varphi(x, y, z)$, 其中 f, φ 可微. 记

$$X = (u^2 - x^2, u^2 - y^2, u^2 - z^2).$$

若 $xyz u \neq 0$ 且 $\partial_1 f(X) + \partial_2 f(X) + \partial_3 f(X) \neq 0$,

求证: $\frac{u'_x}{x} + \frac{u'_y}{y} + \frac{u'_z}{z} = \frac{1}{u}$.

证明: 由于 $f(u^2 - x^2, u^2 - y^2, u^2 - z^2) = 0$, 故

$$\begin{aligned} \partial_1 f(X) \frac{\partial}{\partial x}(u^2 - x^2) + \partial_2 f(X) \frac{\partial}{\partial x}(u^2 - y^2) \\ + \partial_3 f(X) \frac{\partial}{\partial x}(u^2 - z^2) = 0, \end{aligned}$$

也即我们有

$$\partial_1 f(X)(2uu'_x - 2x) + \partial_2 f(X)(2uu'_x) + \partial_3 f(X)(2uu'_x) = 0,$$

$$\text{故 } \frac{u'_x}{x} (\partial_1 f(X) + \partial_2 f(X) + \partial_3 f(X)) = \frac{1}{u} \partial_1 f(X).$$

由对称性立刻可得

$$\frac{u'_y}{y}(\partial_1 f(X) + \partial_2 f(X) + \partial_3 f(X)) = \frac{1}{u} \partial_2 f(X),$$
$$\frac{u'_z}{z}(\partial_1 f(X) + \partial_2 f(X) + \partial_3 f(X)) = \frac{1}{u} \partial_3 f(X).$$

于是我们就有

$$\frac{u'_x}{x} + \frac{u'_y}{y} + \frac{u'_z}{z} = \frac{1}{u}.$$

例 29. $\forall (x, y) \in \mathbb{R}^2$, 定义 $f(x, y) = \sqrt{x^2 + y^4}$.

研究 f 在原点的连续性, 可导性以及可微性.

解: 由于 f 为初等函数, 因此在原点连续.

与此同时, 由于极限

$$\lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{|x|}{x}$$

不存在, 故 f 在原点处没有关于第一个变量的偏导数, 进而可知 f 在原点处不可微.

例 30. 假设 $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ 为二阶连续可导函数, 而隐函数 $z = z(x, y)$ 可由方程 $x + y = f(x, z)$ 确定, 其中 $\partial_2 f(x, z) \neq 0$. 计算 $\frac{\partial^2 z}{\partial x \partial y}$.

解: 将方程两边分别对 x, y 求偏导可得

$$1 = \partial_1 f(x, z) + \partial_2 f(x, z) \frac{\partial z}{\partial x}, \quad 1 = \partial_2 f(x, z) \frac{\partial z}{\partial y},$$

由此我们立刻可知

$$\frac{\partial z}{\partial x} = \frac{1 - \partial_1 f(x, z)}{\partial_2 f(x, z)}, \quad \frac{\partial z}{\partial y} = \frac{1}{\partial_2 f(x, z)}.$$

于是我们有

$$\begin{aligned}\frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{1}{\partial_2 f(x, z)} \right) = -\frac{1}{(\partial_2 f(x, z))^2} \frac{\partial}{\partial x} (\partial_2 f(x, z)) \\&= -\frac{1}{(\partial_2 f(x, z))^2} \left(\partial_{12} f(x, z) + \partial_{22} f(x, z) \frac{\partial z}{\partial x} \right) \\&= -\frac{1}{(\partial_2 f(x, z))^2} \left(\partial_{12} f(x, z) + \partial_{22} f(x, z) \cdot \frac{1 - \partial_1 f(x, z)}{\partial_2 f(x, z)} \right) \\&= \frac{\partial_{22} f(x, z) \cdot \partial_1 f(x, z) - \partial_{12} f(x, z) \cdot \partial_2 f(x, z) - \partial_{22} f(x, z)}{(\partial_2 f(x, z))^3}.\end{aligned}$$

例 31. 设 $q \in (0, 1)$ 而向量值函数 $\vec{f}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ 使得 $\forall X, Y \in \mathbb{R}^n$, 均有

$$\|\vec{f}(X) - \vec{f}(Y)\| \leq q\|X - Y\|.$$

求证: $\exists! A \in \mathbb{R}^n$ 使得 $\vec{f}(A) = A$.

证明: 唯一性.

设 $\exists A_1, A_2 \in \mathbb{R}^n$ 使 $\vec{f}(A_1) = A_1$,
 $\vec{f}(A_2) = A_2$, 则我们有

$$\|A_1 - A_2\| = \|\vec{f}(A_1) - \vec{f}(A_2)\| \leq q\|A_1 - A_2\|,$$

但 $0 < q < 1$, 故 $A_1 = A_2$.

存在性. 任取 $X_1 \in \mathbb{R}^n$. $\forall k \geq 1$, 我们递归定义 $X_{k+1} = \vec{f}(X_k)$. 则 $\forall k \geq 2$, 我们有

$$\|X_{k+1} - X_k\| = \|\vec{f}(X_k) - \vec{f}(X_{k-1})\| \leq q\|X_k - X_{k-1}\|,$$

于是 $\forall k \geq 1$, 我们有 $\|X_{k+1} - X_k\| \leq q^{k-1}\|X_2 - X_1\|$, 进而得 $\{X_k\}$ 为 **Cauchy** 列, 从而收敛. 将其极限记为 A . 那么 $\forall k \geq 1$, 我们有

$$\|X_{k+1} - \vec{f}(A)\| = \|\vec{f}(X_k) - \vec{f}(A)\| \leq q\|X_k - A\|.$$

由夹逼原理可知 $\vec{f}(A)$ 为 $\{X_{k+1}\}$ 的极限, 从而也为 $\{X_k\}$ 的极限, 故 $\vec{f}(A) = A$.

例 32. 求函数 $f(x, y) = \begin{cases} \frac{x^3 - y^3}{x^2 + y^2}, & x^2 + y^2 \neq 0 \\ 0, & x^2 + y^2 = 0 \end{cases}$
在原点处的偏导数 $f'_x(0, 0)$, $f'_y(0, 0)$, 并考察 f
在原点处的连续性和可微性.

解: 由偏导数的定义可知

$$f'_x(0, 0) = \lim_{x \rightarrow 0} \frac{x}{x} = 1, \quad f'_y(0, 0) = \lim_{y \rightarrow 0} \frac{-y}{y} = -1.$$

$\forall (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$, 我们有

$$0 \leq |f(x, y)| \leq \frac{|x|^3 + |y|^3}{x^2 + y^2} \leq \frac{2(x^2 + y^2)^{\frac{3}{2}}}{x^2 + y^2} = 2\sqrt{x^2 + y^2},$$

于是由夹逼原理可知

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0 = f(0,0),$$

从而 f 在原点处连续. 下证 f 在原点处不可微.
用反证法, 假设 f 在原点处可微, 则

$$\begin{aligned} 0 &= \lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - (x-y)}{\sqrt{x^2 + y^2}} \\ &= \lim_{x \rightarrow 0^+} \frac{f(x, -x) - (x+x)}{\sqrt{x^2 + x^2}} = \lim_{x \rightarrow 0^+} \frac{-x}{\sqrt{2}x} = -\frac{\sqrt{2}}{2}, \end{aligned}$$

矛盾! 故 f 在原点处不可微.

例 33. 假设 φ 为二阶连续可微, 而 $z = z(x, y)$ 是由函数方程 $x^3 + y^3 + z^3 = \varphi(z)$ 确定的隐函数, 求 $\frac{\partial^2 z}{\partial x \partial y}$, 并说明隐函数存在的条件.

解: 定义 $F(x, y, z) = x^3 + y^3 + z^3 - \varphi(z)$, 则 F 为二阶连续可微并且 $\frac{\partial F}{\partial z}(x, y, z) = 3z^2 - \varphi'(z)$, 则当 $3z^2 - \varphi'(z) \neq 0$ 时, 由方程 $F(x, y, z) = 0$ 可确定隐函数 $z = z(x, y)$. 此时我们还有

$$\frac{\partial z}{\partial x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}} = \frac{3x^2}{\varphi'(z) - 3z^2}, \quad \frac{\partial z}{\partial y} = -\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}} = \frac{3y^2}{\varphi'(z) - 3z^2},$$

由此立刻可得

$$\begin{aligned}\frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{3y^2}{\varphi'(z) - 3z^2} \right) \\ &= -\frac{3y^2}{(\varphi'(z) - 3z^2)^2} (\varphi''(z) - 6z) \frac{\partial z}{\partial x} \\ &= \frac{9x^2 y^2 (6z - \varphi''(z))}{(\varphi'(z) - 3z^2)^3}.\end{aligned}$$

例 34. 求函数 $z = \frac{\sin x}{1 - \sin y}$ 在 origin $(0, 0)$ 处带二阶 Peano 余项的 Taylor 展式.

解: 当 $(x, y) \rightarrow (0, 0)$ 时, 我们有

$$\begin{aligned}\frac{\sin x}{1 - \sin y} &= \sin x (1 + \sin y + o(\sin y)) \\ &= (x + o(x^2))(1 + y + o(y)) \\ &= x + xy + xo(y) + (1 + y)o(x^2) \\ &= x + xy + o(x^2 + y^2).\end{aligned}$$

例 35. 设 $f(x, y) = \begin{cases} x - y + \frac{xy^3}{x^2 + y^4}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$

求证: 函数 f 在 origin 处连续, 沿任意方向的方向导数都存在, 但不可微.

证明: (1) $\forall (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$, 我们有

$$\begin{aligned} |f(x, y) - f(0, 0)| &= \left| x - y + \frac{xy^3}{x^2 + y^4} \right| \\ &\leq |x| + |y| + \frac{|xy^2| \cdot |y|}{x^2 + y^4} \leq |x| + |y| + \frac{1}{2}|y|. \end{aligned}$$

于是由夹逼原理可知函数 f 在 origin 连续.

(2) 固定 $\vec{\ell}^0 = (\cos \theta, \sin \theta)$. 由定义可知

$$\begin{aligned}\frac{\partial f}{\partial \vec{\ell}^0}(0, 0) &= \lim_{h \rightarrow 0^+} \frac{f(h\vec{\ell}^0) - f(0, 0)}{h} \\&= \lim_{h \rightarrow 0^+} \left(\cos \theta - \sin \theta + \frac{h^4(\cos \theta) \sin^3 \theta}{h(h^2 \cos^2 \theta + h^4 \sin^4 \theta)} \right) \\&= \cos \theta - \sin \theta.\end{aligned}$$

故 f 在原点处沿任意方向的方向导数存在.

(3) 用反证法, 假设 f 在原点可微. 由定义可得

$\frac{\partial f}{\partial x}(0, 0)=1, \frac{\partial f}{\partial y}(0, 0)=-1$. 由复合函数极限法则,

$$\begin{aligned} 0 &= \lim_{(x,y) \rightarrow (0,0)} \frac{f(x, y) - f(0, 0) - \frac{\partial f}{\partial x}(0, 0)x - \frac{\partial f}{\partial y}(0, 0)y}{\sqrt{x^2 + y^2}} \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{xy^3}{(x^2 + y^4)\sqrt{x^2 + y^2}} \\ &= \lim_{y \rightarrow 0^+} \frac{y^2 \cdot y^3}{(y^4 + y^4)\sqrt{y^4 + y^2}} = \frac{1}{2}. \end{aligned}$$

矛盾! 故 f 在原点处不可微.

例 36. 设隐函数 $z = z(x, y)$ 由方程 $x = u + v$, $y = u^2 + v^2$, $z = u^3 + v^3$ 确定. 求 $\frac{\partial^2 z}{\partial x^2}$.

解: 方法 1. 由题设可知

$$1 = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x}, \quad 0 = 2u \frac{\partial u}{\partial x} + 2v \frac{\partial v}{\partial x}.$$

由此我们可立刻导出

$$\frac{\partial u}{\partial x} = \frac{v}{v - u}, \quad \frac{\partial v}{\partial x} = \frac{u}{u - v},$$

于是 $\frac{\partial z}{\partial x} = 3u^2 \frac{\partial u}{\partial x} + 3v^2 \frac{\partial v}{\partial x} = -3uv$, 进而可得

$$\frac{\partial^2 z}{\partial x^2} = -3v \frac{\partial u}{\partial x} - 3u \frac{\partial v}{\partial x} = -3(u + v) = -3x.$$

方法 2. 由题设可知 $2uv = x^2 - y$, 从而

$$\begin{aligned} z &= (u + v)(u^2 - uv + v^2) \\ &= x\left(y - \frac{1}{2}(x^2 - y)\right) \\ &= \frac{3}{2}xy - \frac{1}{2}x^3, \end{aligned}$$

由此我们立刻可得

$$\frac{\partial z}{\partial x} = \frac{3}{2}y - \frac{3}{2}x^2, \quad \frac{\partial^2 z}{\partial x^2} = -3x.$$

例 37. $\forall x, y, z > 0$, 定义

$$f(x, y, z) = \log x + 2 \log y + 3 \log z.$$

求 f 在球面 $x^2 + y^2 + z^2 = 6r^2$ ($r > 0$) 上的最大值, 并证明 $\forall a, b, c > 0$, 均有

$$ab^2c^3 \leq 108 \left(\frac{a+b+c}{6} \right)^6.$$

解: 令 $S = \{(x, y, z) \mid x, y, z > 0, x^2 + y^2 + z^2 = 6r^2\}$, 则 S 为二维曲面.

固定 $P^* \in S$. 注意到

$$\lim_{u \rightarrow 0^+} \log u = -\infty,$$

而 $\forall (x, y, z) \in S$, 我们有

$$\begin{aligned} f(x, y, z) &= \log x + 2 \log y + 3 \log z \\ &\leq \log x + 2 \log(\sqrt{6}r) + 3 \log(\sqrt{6}r) \\ &= \log x + 5 \log(\sqrt{6}r), \end{aligned}$$

同理, 我们也有

$$\begin{aligned} f(x, y, z) &\leq 2 \log y + 4 \log(\sqrt{6}r), \\ f(x, y, z) &\leq 3 \log z + 3 \log(\sqrt{6}r). \end{aligned}$$

由此知 $\exists \varepsilon > 0$ 使得 $\forall (x, y, z) \in S$, 当 $0 < x < \varepsilon$ 或者 $0 < y < \varepsilon$ 或者 $0 < z < \varepsilon$ 时, 我们总会有 $f(x, y, z) < f(P^*)$. 定义

$$S_\varepsilon = \{(x, y, z) \mid x, y, z \geq \varepsilon, x^2 + y^2 + z^2 = 6r^2\}.$$

则 S_ε 为有界闭集, 并且 $P^* \in S_\varepsilon$. 而 f 为连续函数, 于是它在 S_ε 上有最大值. 由前面的讨论可知, 该最大值也是 f 在 S 上的最大值. 我们将相应的最大值点记作 (x_0, y_0, z_0) .

$\forall x, y, z > 0$ 以及 $\lambda \in \mathbb{R}$, 定义

$$L(x, y, z, \lambda) = \log x + 2 \log y + 3 \log z \\ + \lambda(x^2 + y^2 + z^2 - 6r^2).$$

由 Lagrange 乘数法可知, $\exists \lambda \in \mathbb{R}$ 使得

$$0 = \frac{\partial L}{\partial x}(x_0, y_0, z_0, \lambda) = \frac{1}{x_0} + 2\lambda x_0,$$

$$0 = \frac{\partial L}{\partial y}(x_0, y_0, z_0, \lambda) = \frac{2}{y_0} + 2\lambda y_0,$$

$$0 = \frac{\partial L}{\partial z}(x_0, y_0, z_0, \lambda) = \frac{3}{z_0} + 2\lambda z_0,$$

$$0 = \frac{\partial L}{\partial \lambda}(x_0, y_0, z_0, \lambda) = x_0^2 + y_0^2 + z_0^2 - 6r^2.$$

于是我们有

$$x_0 = \frac{1}{\sqrt{-2\lambda}}, \quad y_0 = \frac{1}{\sqrt{-\lambda}}, \quad z_0 = \frac{\sqrt{3}}{\sqrt{-2\lambda}},$$
$$-\frac{1}{2\lambda} - \frac{1}{\lambda} - \frac{3}{2\lambda} - 6r^2 = 0.$$

从而 $\lambda = -\frac{1}{2r^2}$, 进而可知 f 在 D 上的最大值点为 $(r, \sqrt{2}r, \sqrt{3}r)$, 相应的最大值为

$$f(r, \sqrt{2}r, \sqrt{3}r) = 6 \log r + \log 2 + \frac{3}{2} \log 3.$$

$\forall a, b, c > 0$, 我们令

$$r = \sqrt{\frac{1}{6}(a + b + c)},$$

则 $(\sqrt{a})^2 + (\sqrt{b})^2 + (\sqrt{c})^2 = 6r^2$, 从而我们有

$$\begin{aligned} \log \sqrt{a} + 2 \log \sqrt{b} + 3 \log \sqrt{c} \\ \leq 6 \log r + \log 2 + \frac{3}{2} \log 3. \end{aligned}$$

由此我们立刻可得

$$ab^2c^3 \leq 108r^{12} = 108 \left(\frac{a + b + c}{6} \right)^6.$$

例 38. 设 $D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$, $f \in \mathcal{C}(D)$ 在 D 的内部可微使得 $\forall (x, y) \in D, |f(x, y)| \leq 1$. 求证: $\exists (x_0, y_0) \in \overset{\circ}{D}$ 使得我们有

$$\left(\frac{\partial f}{\partial x}(x_0, y_0)\right)^2 + \left(\frac{\partial f}{\partial y}(x_0, y_0)\right)^2 \leq 16.$$

证明: $\forall (x, y) \in D$, 令 $g(x, y) = f(x, y) + 2(x^2 + y^2)$. 则 $g \in \mathcal{C}(D)$ 在 D 的内部可微且 $\forall (x, y) \in \partial D$, $g(x, y) = f(x, y) + 2 \geq 1$. 又 $g(0, 0) = f(0, 0) \leq 1$, 且 $g \in \mathcal{C}(D)$, 于是我们由最值定理立刻可得知,

$\exists (x_0, y_0) \in \overset{\circ}{D}$ 使得函数 g 在该点处取到最小值,

则该点也为 g 的极小值点, 而 g 在该点处可微, 于是点 (x_0, y_0) 为 g 的驻点, 也即

$$\begin{aligned} 0 &= \frac{\partial g}{\partial x}(x_0, y_0) = \frac{\partial f}{\partial x}(x_0, y_0) + 4x_0, \\ 0 &= \frac{\partial g}{\partial y}(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0) + 4y_0, \end{aligned}$$

由此我们立刻可得

$$\left(\frac{\partial f}{\partial x}(x_0, y_0)\right)^2 + \left(\frac{\partial f}{\partial y}(x_0, y_0)\right)^2 = 16(x_0^2 + y_0^2) \leq 16.$$

谢谢大家!